Does DTW Improve Regime-Aware Portfolio Construction?

Mid Semester Report

By

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October 2026

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Acknowledgements

I would like to thank Utkarsh Kumar for guiding this project. The expanding-window design and the two-period, multi-horizon extension described in this report were both suggested during our discussions.

1 Introduction

1.1 Motivation

Mean-variance portfolio construction needs an estimate of the covariance matrix of stock returns [1]. The sample covariance only compares returns that occur on the same day. This misses a well-documented feature of equity markets: some stocks react to common information later than others. Lo and MacKinlay [2] found that returns of large stocks lead those of small stocks, and Hou [3] linked this lead-lag effect to slow diffusion of industry information. If stock A tends to follow stock B with a delay of a few days, their same-day covariance understates how closely the two are related.

I chose Dynamic Time Warping (DTW) because it compares two series after allowing them to be shifted and stretched in time [4, 5]. Unlike a lagged correlation, it does not require a single fixed lag to be chosen in advance, and the alignment can change within the sample. The idea is to use a DTW-based similarity in place of same-day correlation and check whether portfolios built on it perform better out of sample.

The second part of the design comes from the fact that dependence between stocks changes with market conditions. Regime-switching models [6] and their use in asset allocation [7, 8] show that covariance estimated separately for calm and stressed periods can lead to different portfolios than a single unconditional estimate. For this reason I detect market regimes with a Hidden Markov Model (HMM) [9] and estimate both covariance matrices separately inside each regime.

1.2 Related Work and Gap

The papers cited above treat these ideas separately. The lead-lag studies [2, 3] measure delayed reactions using lagged cross-correlations, but they do not turn them into a covariance input for optimization. Regime-based allocation [7, 8] conditions the covariance on the regime but still uses same-day co-movement. Covariance improvements such as shrinkage [10] reduce estimation error but do not address timing. I have not yet carried out a systematic search for work that uses DTW-derived similarity as a regime-conditional covariance input. Whether this combination is new, and how it compares with existing DTW-based clustering or portfolio methods, is the first task planned for the second half of the semester (Section 8).

2 Objective and Scope

The research question is:

> *Within HMM-detected market regimes, does a DTW-based covariance matrix improve out-of-sample portfolio performance compared to the standard sample covariance? If it does, are the gains statistically significant and robust across training windows and time periods?*

Performance is measured by the annualized Sharpe ratio [11]. Both methods use the same regimes, the same mean-return estimates and the same constraints. Apart from the covariance matrix itself, the only difference is that the DTW matrix can include only stocks that are present in every episode of a regime (Section 3.4).

The study uses three Indian equity universes: Nifty 50, Nifty Bank (14 stocks) and Nifty Midcap 100. Daily adjusted closing prices for the constituents and for the corresponding index (Nifty 50, Nifty Bank and Nifty Midcap 100) were downloaded from Yahoo Finance, covering January 2008 to August 2026. The constituent lists are the current ones. For the 2008–2019 period this means only stocks that were already listed by the start of 2009 are kept: 43 of 50 for Nifty 50 and 12 of 14 for Nifty Bank. This introduces survivorship bias, which is discussed in Section 7.

3 Methodology

3.1 Regime Detection

Regimes are detected from the index of each universe. With I_t the index level and $r_t = \ln(I_t/I_{t-1})$, the feature vector on day t is $x_t = (r_t,\ \sigma_t,\ DD_t,\ m_t)$, where

$$\sigma_t = \mathrm{sd}(r_{t-19}, \ldots, r_t), \qquad DD_t = C_t - \max_{t-59 \le u \le t} C_u, \quad C_t = \textstyle\sum_{u \le t} r_u,$$
$$m_t = \tfrac{1}{5} \textstyle\sum_{u=t-4}^{t} r_u.$$

So σ_t is the 20-day volatility, DD_t is the 60-day drawdown measured on cumulative log returns, and m_t is the 5-day average return. A Gaussian HMM with four states and full covariance matrices is fitted on the training features using the EM algorithm (at most 5000 iterations, tolerance 10^{-4}). EM can stop at a poor local optimum, so the model is fitted with five random seeds and the fit with the highest training log-likelihood is kept. Seeds where a state covariance became non-positive-definite during fitting are skipped.

Merging four states into three regimes. HMM state numbers are arbitrary and change between fits, so the mapping is rebuilt in every fold. The four states are ranked by the mean of r_t over the training days assigned to them. The two lowest-return states are labelled Bear, the third Sideways and the highest Bull. If a fold uses fewer than four states, the remaining ones are mapped in the same order.

Smoothing. Raw HMM labels can switch for a single day. Each label sequence is smoothed with a trailing 60-day mode filter, $\tilde{y}_t = \mathrm{mode}(y_{t-60}, \ldots, y_{t-1})$, which only uses earlier days (the first 60 days keep their raw labels). The cost is a delay: throughout the sharpest part of the March 2020 crash, the smoothed label was still Sideways.

3.2 Regime Episodes

An episode is a maximal run of consecutive days with the same smoothed regime. Two runs of the same regime separated by an interruption of at most 3 days are joined, so that a short flip inside a long Bull run does not split it in two. Episodes shorter than 10 days are dropped, since there is too little data in them to estimate co-movement. The episodes of regime R in a training window are $\mathcal{E}_R = \{e_1, \ldots, e_K\}$.

3.3 Standard Covariance

For regime R, the daily log returns of all training days that belong to its episodes are pooled. Stocks with at least 30 valid observations are kept, and Σ_R^{std} is the pairwise sample covariance of these returns (at least 30 overlapping observations per pair).

3.4 DTW-Based Covariance

Distance per episode. Within each episode, stocks with at least 90% non-missing days are kept and days with any missing value are dropped. For two return series x and y, DTW finds

the alignment with the lowest total cost through the recursion

$$\gamma(p,q) = |x_p - y_q| + \min\{\gamma(p-1,q),\ \gamma(p,q-1),\ \gamma(p-1,q-1)\},$$

and the distance is the value in the final cell. In the code this is computed with FastDTW [12] (default radius 1), which approximates the exact DTW distance in linear time. This was needed because each fold requires a distance for every stock pair in every episode.

Averaging across episodes. DTW is run inside each episode and not on the regime's days joined end to end. Joining them would let the alignment match the end of one episode with the start of another from a different year. For stocks i and j, the episode distances are averaged over all K episodes of the regime, using the stocks that are present in every episode:

$$\bar{D}_{ij} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{DTW}\big(r_i^{(k)},\ r_j^{(k)}\big).$$

Because the DTW distance is a sum along the alignment path, longer episodes produce larger distances and carry more weight in this average. Normalizing by episode length is one of the planned robustness checks.

Gaussian kernel and bandwidth. The averaged distances are converted to similarities with

$$S_{ij} = \exp\left(-\frac{\bar{D}_{ij}^2}{2h^2}\right), \qquad h = c \cdot \mathrm{median}\{\bar{D}_{ij} : \bar{D}_{ij} > 0\}.$$

Expressing h relative to the median distance makes the bandwidth comparable across regimes and universes. The factor c was chosen with a sweep over $c \in \{0.1, 0.2, 0.3, 0.5, 0.75, 1.0, 1.5\}$ on a training window (2015–2021 for the single-split and expanding-window studies, and the middle fold's training window, 2008–2015 or 2020–2022, for the two-period studies), recording for each regime the condition number of the resulting covariance matrix and the standard deviation of the off-diagonal similarities. A small c makes all off-diagonal similarities close to zero, so the matrix becomes nearly diagonal. A large c makes them all close to one, so the matrix becomes nearly singular. The rule was to keep only values whose median off-diagonal standard deviation is at least 0.05, and then pick the one with the lowest median condition number. This rule gave $c = 0.5$ in every universe and period. At $c = 0.5$ the condition numbers ranged from single digits to the low hundreds, while for $c \geq 0.75$ they were usually in the hundreds or higher and in some windows exceeded 10^9. The choice depends only on the numerical behaviour of the matrix, not on portfolio returns.

Eigenvalue correction. S is symmetric but not guaranteed to be positive semi-definite. With the eigendecomposition $S = Q\Lambda Q^\top$, eigenvalues below 10^{-8} are set to 10^{-8}:

$$\tilde{S} = Q \max(\Lambda, 10^{-8} I)\, Q^\top.$$

The diagonal of $\tilde{S}$ is not rescaled afterwards, so it is close to, but not exactly, a correlation matrix.

Volatility scaling. With σ_i the standard deviation of stock i's daily log returns over the regime's training days,

$$\Sigma_R^{\mathrm{DTW}} = D_\sigma\, \tilde{S}\, D_\sigma, \qquad D_\sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_N).$$

3.5 Portfolio Optimization

For each regime and each covariance matrix, the long-only maximum Sharpe ratio portfolio is computed:

$$w_R = \arg\max_w \ \frac{w^\top \mu_R}{\sqrt{w^\top \Sigma_R\, w}} \quad \text{s.t.} \quad \sum_i w_i = 1, \quad 0 \leq w_i \leq 0.4,$$

where μ_R is the vector of mean daily log returns over the regime's training days. No risk-free rate is subtracted. The problem is solved with SLSQP starting from equal weights. The 40% cap prevents the optimizer from putting nearly all of the weight on one or two stocks, which unconstrained max-Sharpe solutions tend to do. The weights are then frozen for the whole test year.

On each test day t, the portfolio of the regime decoded for that day is held, so the daily return is $R_t = w_{\hat{s}_t}^{\top} r_t$, where r_t is the vector of stock log returns and missing returns are set to zero. If no weights could be built for a regime in a fold, test days in that regime are skipped for both methods instead of being filled with a default portfolio.

3.6 Leakage Controls

Two look-ahead errors were found and fixed during development:

- **Centered smoothing.** An earlier version used a centered rolling window to smooth labels, which uses future days. Any smoothing that touches test data is now trailing only, as in Section 3.1.
- **Whole-sequence decoding.** Calling the Viterbi decoder once on the full test sequence lets later test days influence the label on earlier days. Test labels are now decoded online: the label for day t is the last state of the Viterbi path computed on the test features observed up to t,

$$\hat{s}_t = \Big[\arg\max_{s_\tau, \ldots, s_t} P(s_\tau, \ldots, s_t \mid x_\tau, \ldots, x_t) \Big]_t ,$$

  where τ is the first test day. In the single-split study this fix changed the standard portfolio's Sharpe ratio from 0.508 to 0.541, which shows that the leak was small enough to go unnoticed.

In addition, the HMM, the regime map, the episodes, both covariance matrices, the mean returns and the weights are all estimated from training data only.

4 Experimental Design

4.1 Study Designs

The 16 significance tests come from three study designs that were run one after another, each addressing a weakness of the previous one (Table 1).

Design	Universe	Training	Testing	Horizons	Tests
Single split	Nifty 50	2015–2021	2022–Aug 2026	1d	1
Expanding window	N50, Bank, Midcap	2015 to $y-1$	y = 2020–2025	1d	3
Two-period A	N50 (43), Bank (12)	2008 to $y-1$	y = 2013–2019	1, 5, 22d	6
Two-period B	N50 (50), Bank (14)	2020 to $y-1$	y = 2021–2025	1, 5, 22d	6

Table 1: Study designs. Each test year is a separate fold with its own HMM, covariance matrices and weights. Results are pooled over folds.

The **single split** gives only one out-of-sample period, so it cannot separate a real effect from a lucky test period. The **expanding-window** design produces six yearly test folds instead, while still growing the training set over time. The **two-period** design then splits the history into two eras that do not share any data, and evaluates three return horizons. In Period B, the requirement is *strict separation*: no price data from before 1 January 2020 enters any computation, including the warm-up of the rolling features. This is what distinguishes it from

the expanding-window study, which tests almost the same years (2020–2025) but trains from 2015. As a consequence, the first 60 or so trading days of 2020 are used up by the rolling windows, and fold 1 of Period B trains on only about 190 days.

4.2 Why 16 Tests and Not 18

The 16 tests are not a full grid. They are every DTW-versus-standard significance test run in the project: 1 single split, 3 expanding-window universes and 12 two-period tests (2 universes $\times$ 2 periods $\times$ 3 horizons). A complete two-period grid over three universes would have 18 tests. Midcap 100 was left out of the two-period design on purpose: midcap membership is defined by relative market capitalization and changes constantly, so today's Midcap 100 list is not a reasonable stand-in for the midcap stocks of 2008. Large banks and large-cap stocks change much less over this period, so the current Nifty 50 and Nifty Bank lists are more defensible for Period A. Three other analyses were not counted as tests: the Midcap result recomputed without 2023 (a sensitivity check on data already in test 4), an abandoned 2015–2019 training split (stopped because the Sideways regime had only 58 training days for 45 stocks), and the dry runs used to check the code.

4.3 Horizons and Overlapping Returns

For the 5-day and 22-day horizons, the daily portfolio log returns are summed over consecutive *non-overlapping* blocks of h days, and any days left over at the end are dropped. Overlapping windows would create artificial autocorrelation and make the effective sample look larger than it is. Non-overlapping blocks avoid this, but leave few observations: Period B has 247 five-day and 56 twenty-two-day returns. The Sharpe ratio at horizon h is the mean over the standard deviation of these returns, multiplied by $\sqrt{252/h}$.

5 Results

Table 2 lists all 16 tests.

#	Test	Standard	DTW	Difference	p (two-sided)
1	Single split, Nifty 50, 1d	0.541	0.669	+0.128	0.600
2	Expanding window, Nifty 50, 1d	1.309	1.040	−0.269	0.110
3	Expanding window, Nifty Bank, 1d	0.396	0.345	−0.051	0.870
4	Expanding window, Midcap 100, 1d	0.807	1.290	+0.483	0.164
5	Period A, Nifty 50, 1d	1.190	1.214	+0.024	0.886
6	Period A, Nifty 50, 5d	1.095	1.134	+0.039	0.624
7	Period A, Nifty 50, 22d	1.117	1.133	+0.016	0.320
8	Period B, Nifty 50, 1d	1.063	1.184	+0.121	0.586
9	Period B, Nifty 50, 5d	1.057	1.239	+0.182	0.385
10	Period B, Nifty 50, 22d	1.007	1.180	+0.173	0.192
11	Period A, Nifty Bank, 1d	0.821	0.699	−0.122	0.164
12	Period A, Nifty Bank, 5d	0.793	0.668	−0.125	0.049
13	Period A, Nifty Bank, 22d	0.851	0.708	−0.143	0.108
14	Period B, Nifty Bank, 1d	0.908	0.961	+0.053	0.723
15	Period B, Nifty Bank, 5d	0.932	0.990	+0.058	0.991
16	Period B, Nifty Bank, 22d	1.019	1.052	+0.033	0.863

Table 2: Out-of-sample annualized Sharpe ratios for all 16 tests, before transaction costs. Difference is DTW minus Standard.

DTW has the higher Sharpe ratio in 11 of the 16 tests. All 6 strict-separation tests (Period B) favor DTW, and 8 of the 10 tests whose test years are 2020 or later do so. The six pre-2020 tests split by universe: the three Nifty 50 tests favor DTW by very small margins (+0.016 to +0.039), while the three Nifty Bank tests favor the standard covariance (−0.122 to −0.143).

In the single-split study, the regime breakdown shows where the difference comes from. DTW did better in Sideways (1.462 against 1.006) and Bear (0.843 against 0.728) but worse in Bull (−1.195 against −0.680). Bull weights trained on 2015–2021 did not follow the change in market leadership during 2022–2026, and the DTW matrix appears to have made this worse. Charging 10 basis points per unit of turnover on regime-change days lowers both Sharpe ratios by a similar amount (DTW 0.621, standard 0.488). This is because both portfolios switch on the same 36 days, since they share the same regime sequence.

6 Statistical Significance

6.1 Test Used

Each comparison uses a paired moving-block bootstrap [13] of the Sharpe ratio difference $\Delta = \mathrm{SR}_{\mathrm{DTW}} - \mathrm{SR}_{\mathrm{std}}$. The two return series are aligned on common dates. Each resample draws $\lfloor n/20 \rfloor$ blocks of 20 consecutive observations with uniformly random start points, and the same blocks are applied to both series. This keeps the serial dependence within each block and the correlation between the two strategies, which hold largely the same stocks. With $B = 5000$ resamples and resampled differences Δ_b^*,

$$p_{\mathrm{one}} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\{\Delta_b^* \le 0\}, \qquad p_{\mathrm{two}} = 2\min(p_{\mathrm{one}},\ 1 - p_{\mathrm{one}}),$$

and the 95% confidence interval is given by the 2.5th and 97.5th percentiles of Δ^*. The two-sided value is used because results were interpreted in both directions. It is below 0.05 exactly when the percentile interval excludes zero.

6.2 Assumptions and Limitations of the Test

The test assumes that the return difference is stationary over the pooled test period and that dependence dies out within about 20 observations. Pooling concatenates separate folds, so the first assumption is only approximate. The block size is counted in observations, so at the 22-day horizon one block covers about 440 trading days and each resample contains only 2 (Period B) or 3 (Period A) blocks. The 22-day p-values, and to a lesser extent the 5-day ones (12–17 blocks), therefore have low resolution and should be read with caution. The percentile method also does not resample under the null hypothesis, so p_{two} is an approximate measure, not an exact test.

6.3 Multiple Comparisons

With 16 tests, about one result below 0.05 is expected by chance even if the two methods are identical. Two corrections were applied to the two-sided p-values. Bonferroni requires $p < 0.05/16 = 0.003125$. Benjamini-Hochberg [14] sorts the p-values and finds the largest k with $p_{(k)} \le \frac{k}{16} \cdot 0.05$, which controls the false discovery rate. No test survives either correction. The smallest raw p-value (0.049, Nifty Bank Period A at 5 days) has a Benjamini-Hochberg adjusted value of 0.511, and its direction favors the *standard* covariance.

7 Discussion

The main observation is that the sign of the effect is not stable. DTW wins most comparisons, but most of the margins are small, and the largest positive and negative results both depend on specific design choices.

Training-window design changes the result. The clearest case is Nifty 50. The expanding-window study, trained from 2015 and tested on 2020–2025, favors the standard covariance (-0.269). Period B, trained only on data from 2020 onward and tested on 2021–2025, favors DTW at all three horizons ($+0.121$ to $+0.182$). The test years almost completely overlap and the universe is the same, so the change of sign comes mainly from what the frozen weights were trained on.

The universe also matters. Period A uses the same fold years and the same procedure for Nifty 50 and Nifty Bank, yet the gap is slightly positive for Nifty 50 and consistently negative for Nifty Bank. Training design alone does not explain this.

Some results depend on a single year. The Midcap 100 gain ($+0.483$) is driven mostly by 2023, where DTW reached a Sharpe ratio of 3.19 against -0.33 for the standard portfolio. Without 2023 the gap falls to $+0.091$. In Period B for Nifty 50, DTW loses 2021 and 2024 (-0.858) and wins 2022, 2023 and 2025 ($+0.868$), so removing either 2024 or 2025 would change the pooled result considerably.

Period B is the thinnest design. Its first fold trains on about 190 days. For Nifty 50 only the Bear regime survived episode extraction, so the whole of 2021 was traded with a single Bear portfolio. For Nifty Bank, Bull never appeared and its test days were skipped. Later folds have regimes with fewer than 100 training days. The design where DTW looks best is also the one with the least training data.

Other limitations are survivorship bias from using current constituent lists for 2008–2019, the approximation in FastDTW, and the use of a weighted sum of log returns as the portfolio log return. In addition, transaction costs were only applied in the single-split study, and in the expanding-window studies the bandwidth sweep window (2015–2021) overlaps the first two test years. That sweep uses only the conditioning of the matrix, not returns, but it should still be moved inside each fold's training window.

Overall, the results are directionally consistent with DTW helping in the recent era, but they are not statistically significant and they change with the training window and the universe. DTW cannot be claimed as an improvement on the current evidence.

8 Plan After Midsem

1. **Literature search.** Search Google Scholar and SSRN for combinations of "dynamic time warping" with "covariance", "portfolio" and "lead-lag", and of "regime switching" with "asset allocation". Summarize each relevant paper by method, data and finding, and use this to write the gap statement properly.

2. **Mechanism check.** The code currently discards the FastDTW warping path. I will keep it and compute, for each stock pair and episode, the implied lag as the median of $q - p$ along the path. This will be compared with the lag $k^* = \arg\max_k \rho_{ij}(k)$ of the lagged cross-correlation for $k \in \{-5, \ldots, 5\}$, using the rank correlation across pairs. If DTW is capturing lead-lag, the two should agree. I will also compare FastDTW with exact DTW on a sample of pairs to measure the approximation error.

3. **Robustness of the DTW construction.** Run the full backtests with $c \in \{0.3, 0.5, 0.75\}$, with DTW distances divided by episode length, and with minimum episode lengths of 10, 20 and 40 days.

4. **Regime transitions.** Find the Bull-to-Bear switches in the decoded test labels and compare the cumulative returns of the two portfolios from 20 days before to 20 days after each switch. The number of such events is small, so this will be descriptive.

5. **Costs and metrics.** Apply costs of 0, 10 and 20 basis points per unit of turnover on regime-change days to all 16 tests, and report Sortino and Calmar ratios and maximum drawdown along with Sharpe.

6. **Statistical checks.** Repeat the bootstrap with block sizes of 5, 10 and 20, and use the studentized Sharpe-difference test of Ledoit and Wolf [15] as a second test.

7. **Write-up** of the final methodology and results.

9 Conclusion

The pipeline for regime-conditional DTW covariance is complete and has been evaluated in three study designs, three universes and three horizons, with leakage controls checked during development. DTW had the higher Sharpe ratio in 11 of 16 tests and in all 6 strict-separation tests. However, no test is significant after correcting for multiple comparisons, the only nominally significant result favors the standard covariance, and the sign changes with the training window and the universe. The second half of the project will test whether DTW actually captures lead-lag structure, and whether the results hold under different bandwidths, episode rules, transaction costs and statistical tests.

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